



Experts in Aircraft Video Surveillance

An Introduction





About Us

UK headquarters, global footprint.

Bespoke and “off-the-shelf” CCTV solutions for all aviation applications.

Family-owned British business with a sole focus on aircraft video surveillance.

Active partnerships with over 100 airline customers worldwide, as well as with most Aircraft Lessors, MROs, and VIP aircraft operators. Our products can be found on aircraft in every continent of the world.

AS9100/ISO9001, CAA Part 21 (UK.21G.2205), CAA and EASA Part 145 (UK.145.00613) and FAR145 (K8QY576P) approvals lead the impressive list of accreditations.





About Us

AD Aerospace is a Boeing approved supplier (Supplier Code 694845), a Key Vendor award winner, and approved for production fit Flight Deck Entry Video Surveillance Systems (FDEVSS) on the Boeing 737MAX.

AD Aerospace is also an Airbus approved production partner to the Airbus A220 aircraft production line through a Long-Term Agreement with CMC Electronics in Canada. AD Aerospace also delivers external cameras for the Airbus Helicopters range of helicopters (Airbus Supplier Code 5045106).

AD Aerospace is also an approved supplier to AVIC and COMAC aircraft manufacturers in China for the ARJ21 and C919 aircraft.





About Us

The Group

Mythra Ltd



Design, manufacture and marketing of aircraft video surveillance solutions.



Supplier of complementary aerospace video surveillance equipment and medical equipment included Philips Automated External Defibrillators (AEDs) and Tempus IC2 On-Board Diagnostic Device.

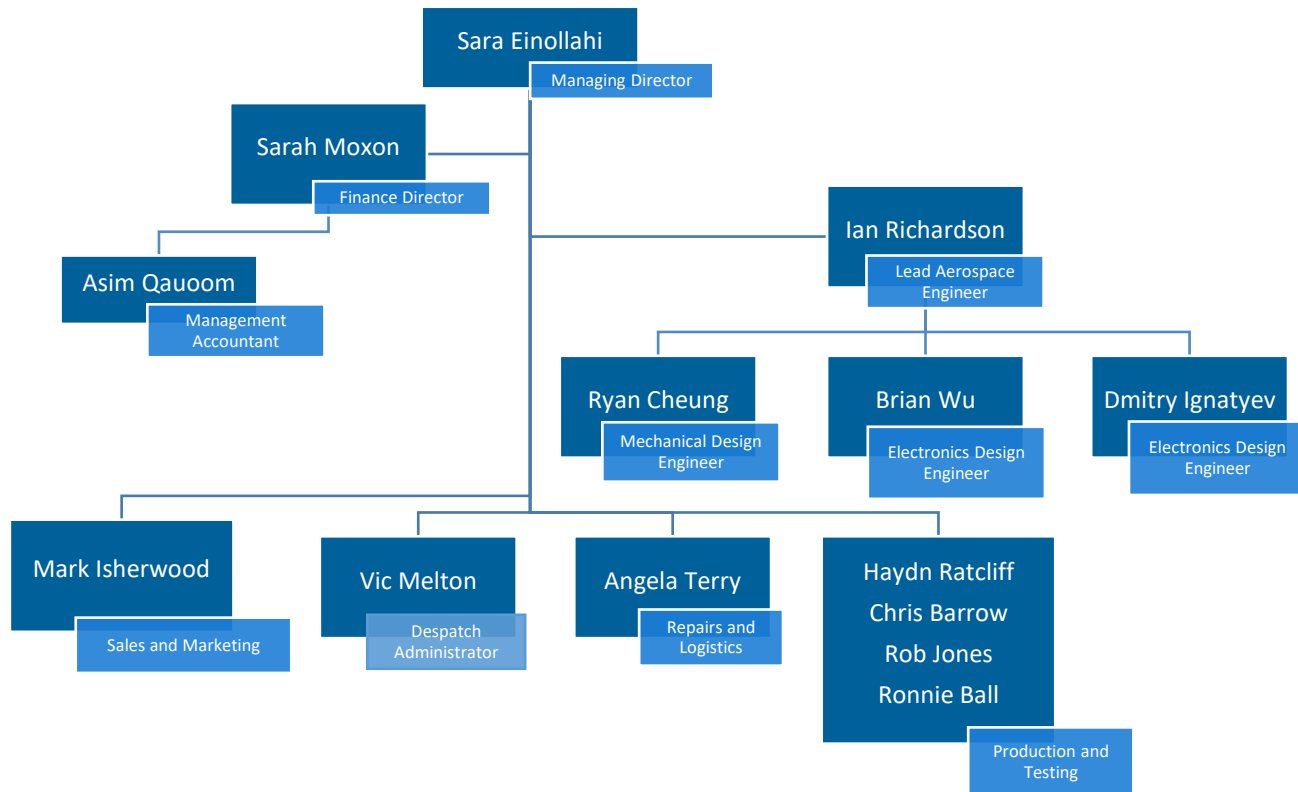


Dedicated Micros is our specialist production company based in Malta.



About Us

The Team



● Line Fit Flight Deck Entry Video Surveillance System (FDEVSS)

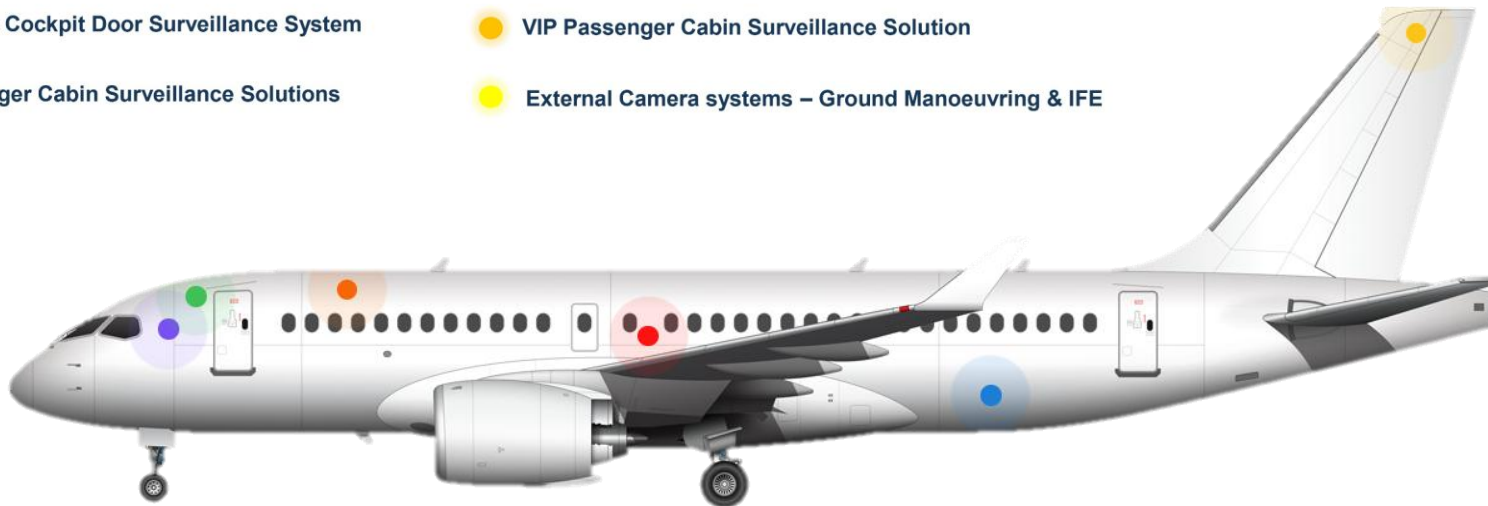
● Retrofit Cockpit Door Surveillance System

● Passenger Cabin Surveillance Solutions

● Cargo Surveillance Solutions – Cargo, baggage & freight bay observation

● VIP Passenger Cabin Surveillance Solution

● External Camera systems – Ground Manoeuvring & IFE



Our Systems



Our Systems

Line-Fit FDEVSS

Flight Deck Entry Video Surveillance (FDEVSS)

Satisfies the requirement for a non-intrusive, non-distracting method of identification by the Captain and First Officer of who is trying to gain access to the Cockpit.

Acts as deterrent to terrorism and air rage and helps to prevent catastrophic loss of aircraft from terrorism or revenge motivation.

Meets EU-OPS Sub Part S, ICAO Annex 6, and EUROCAE ED-123. It also fulfils SAFA requirements.

Video streams directly to Flight Deck Display (MFD, EFB)

Analogue and IP/ Full HD technology.

Available on –
B737MAX, Airbus A220, COMAC C919



Retro-Fit CDSS

Retro-fit Cockpit Door Surveillance (CDSS)

A version of FDEVSS fitted post delivery.

Potential to save on OEM installation costs.

Represents low-cost option to significantly increase levels of aircraft safety and security.

Video streams directly to monitor installed in the Cockpit.

Analogue and IP/ Full HD technology.

Range of EASA and FAA approved STCs / Minor Mods on airframes including:

B737 NG, A320ceo, A320neo, Dash 8 Q400, F70, F100, ATR 72





Our Systems

Passenger Cabin

Passenger cabin cameras used for the prevention of air rage, terrorist crimes, in-flight theft and people trafficking by increasing the information available to pilots and crew.

IATA says that the cost of diverting to offload a disruptive traveller can exceed \$200,000 (£155,000).

Recording option: digital video server can record the entire flight & download to ground via Wi-Fi (subject to availability of airline supplied wireless system interface).

Improved situational awareness enables the Captain to manage aircraft more effectively, advise crew and help avoid unnecessary diversions.

Helps prevent litigation by passengers claiming wrongful identification and helps to establish the exact chain of events leading up to an incident.

Recordings of incidents can be used to help train cabin crew to deal with disruptive passenger behaviour.

IP / Full HD technology with integrated video server to achieve 30 days storage of recorded video.



VIP Cabin – “Direct View”

Comply with CFR 25.785 requirements, designed to provide visibility of a minimum of 50% of passengers in premium class zone to seated cabin crew during Taxi, Take Off and Landing.

Dedicated HD-SDI Camera and Monitor allowing clear visibility of selected areas of passenger cabin not visible from cabin attendant's seat position.

720p HD-SDI unit containing a bullet camera, externally controllable via RS485.

Provides cabin attendants with greater situational awareness of areas of passenger cabin obscured from view when cabin attendants are seated.





Our Systems

Cargo

The system allows for real-time streaming of video to an EFB, MFD or a dedicated LCD display.

The aim of the system is to allow the crew to keep a visual check on the cargo loading, access progress, and to monitor for illegal activity.

Cargo hold and main deck freighter cargo.

Video Server connected to up to 8 (+8) cameras in the aircraft.
Removable CF memory card for easy retrieval of data, or live surveillance.

Analogue or IP / Full HD technology



Ground Manoeuvring

20% of aircraft accidents involve ground manoeuvring, with over half of these involving aircraft on the ground (ICAO, 2018).

Large aircraft present situational awareness challenges to pilots manoeuvring on the ground where wingtips are not visible from the cockpit. This increases the risk of collision with airport ground equipment or structures.

Pilots have had trouble in evaluating the position of the outer main landing gear during taxi, with increased risk of the outer main gear running over the edge of the taxiway whilst conducting turning manoeuvres.

Our system provides the pilot with a clear view of the 'Blind Spots' directly to the EFB or MFD, and video is recorded.

'Blind Spot' visibility is achieved with two cameras: one camera fitted to the left, and one to the right tips of the horizontal stabilizer, for situational awareness.





Our Systems

Helicopter Cameras

Traditional Analogue
HD-SDI - Full HD
IP 4K & Integrated Server

Existing bespoke installations for the following use cases:

Oil & Gas
Search & Rescue
Firefighting
Cargo Hook / Winch Monitoring
Law Enforcement
Engine & Gearbox real-time monitoring
Utilities
Special Mission
Demonstrators



eVTOL Cameras

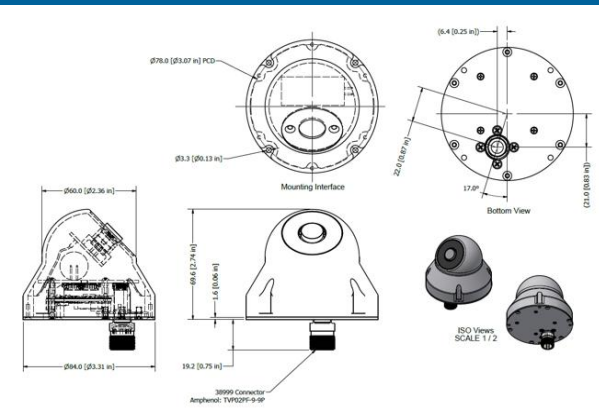
All new 4K IP Camera

Use cases include:

In-Flight Entertainment (IFE)
Pilot Situational Awareness
Vertiport approach & landing
Ground Manoeuvring
Internal Cabin Monitoring

Bespoke housing design to ensure aerodynamic fit with airframe.

FoC units available for testing / demo.







NEW PRODUCT

4K IP VIDEO CAMERA

~ with ~

INTEGRATED RECORDING & STORAGE



FEATURES (1/2)



4K

4K Resolution

In addition to improvement in picture quality and sharpness, 4K resolution enables more detail to be captured, viewed and recorded by the Pilot, First Officer and Crew.

4K resolution will also allow the camera to have a wider range of lens and therefore range of Field of View (FoV) options and low light performance.

The high-resolution camera also offers the potential for digital zoom without sacrificing the quality of the image.

Power over Ethernet (PoE)

The 4K camera will have two options for power; standard 28VDC or Power over Ethernet (PoE).

PoE will enable the camera to only be connected via one wire which will provide both power input and ethernet connection.

The less wiring will reduce the weight of the system, reduce the potential points of failure and lower the installation and maintenance times.

Integrated Server (Recording / Storage)

Integrated server capability will:

- Allow recording and storage from the cameras themselves.
- Eliminate requirement for separate server installation.
- Allow for greater storage capacity - each camera can hold up to 1TB of data i.e., a 6-camera system can hold 6TB of data.
- Allow for remote data access/download via Wi-Fi - reduces time during maintenance checks as the units will not have to be accessed physically/manually.
- Reduce weight by a minimum of 2kg (based on weight and additional wiring of the AD Aerospace FV-0591 Video Server).

ONVIF

The 4K camera will be Open Network Video Interface Forum (ONVIF) compliant.

It will therefore be part of the global open standard for the interface of IP avionics including any such Cabin Experience Network (CEN).

ONVIF compliance means the camera can be connected to third party ONVIF equipment / networks and will be adaptable and easily expandable i.e., cameras added to expand coverage.

FEATURES (2/2)

Qualification

The 4K camera will be fully tested and qualified to D0160G, to enable it to be mounted internally but also external to an aircraft fuselage to provide industry leading external views.

AD Aerospace has significant history of successfully qualifying aerospace cameras for both internal and external applications on a range of aircraft programs.

Materials

The optics of the camera are protected with a sapphire glass window which is scratch resistant, and the housing of the camera is manufactured from Titanium alloy.

For external applications, the unit is designed to be impervious to internal condensation “fogging” as well as external icing. This is due to the inclusion of an internally controlled heater localised to the optical surface, and also nitrogen purging.

Adaptability & Bespoke Housings

The design of the internal components of the 4k camera will be fixed in such a manner that will allow the product to have a range of configurations and bespoke housings. The camera will therefore be adaptable to many different use cases and installations.

Housing options will include internal and external housings as well as bespoke housings based on aircraft specifications. E.g., fuselage dimensions/curvature.

AI

The 4K camera processor will have an integrated dedicated AI processor capable of at least 1 TOPS (Trillion Operations per second). This would allow the camera to perform AI processing at the edge directly on the camera processor, without the need to forward the stream to a dedicated server or cloud processing. This will speed up the AI operations and make the system self-contained.

Typical use cases might include facial recognition at the cockpit door, detection of unknown faces and alarms, passenger counting, face mask wearing, suspicious activity, luggage size detection and many more.



4K

EXAMPLES



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USE CASES / INSTALLATION



4K

Internal Surveillance

The 4K camera can be used for any internal camera requirement on-board any aircraft type. Such as:

- Flight Deck Entry / Cockpit Door Surveillance (FDEVSS/ CDSS), with MFD / Flight Deck Display integration.
- VIP Passenger Cabin Surveillance / Direct View – to comply with 14 CFR 25.785 so the seated cabin crew can view at least 50% of the passengers in a premium class cabin zone.
- Passenger cabin and galley video surveillance as a deterrent and solution to unruly passengers.
- Within the cockpit to monitor instruments and work in conjunction with Flight Recorders.

External (Fixed-Wing)

The 4K camera will have configurations designed specifically for installation external to the aircraft fuselage. Designed for use up to 50,000ft, the unit will be impervious to internal condensation “fogging” as well as external icing, due to an internally controlled heater localised to the optical surface, and nitrogen purging.

External use cases for fixed wing aircraft include:

- In-Flight Entertainment (IFE)
- Pilot Situational Awareness
- Ground Manoeuvring to mitigate against ground collisions.
- Aircraft Refuelling

Cargo / Freighter Surveillance

Cargo bay damage and baggage theft continue to prove costly for airlines and aircraft operators. With the new 4k camera system, as the server capability is integrated within the camera units, only cameras will need to be installed within the cargo bay(s) and using motion detection software the cameras can record as soon as movement is detected. Each camera can store detailed, ultra-high-definition video which can be accessed remotely if an incident is recorded.

External (Helicopter / eVTOL)

AD Aerospace currently sells its FV-0315 analogue external camera and FV-0361 HD-SDI external camera to several rotorcraft programs, the 4K camera will be the perfect replacement. Use cases include:

- Cargo Hook / Winch Monitoring
- Oil and Gas surveys
- Border Control / Police surveillance
- Firefighting
- Real time engine and gearbox monitoring

Advanced Air Mobility (AAM) has its own use cases for a 4K camera including IFE and Pilot Situational Awareness.



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